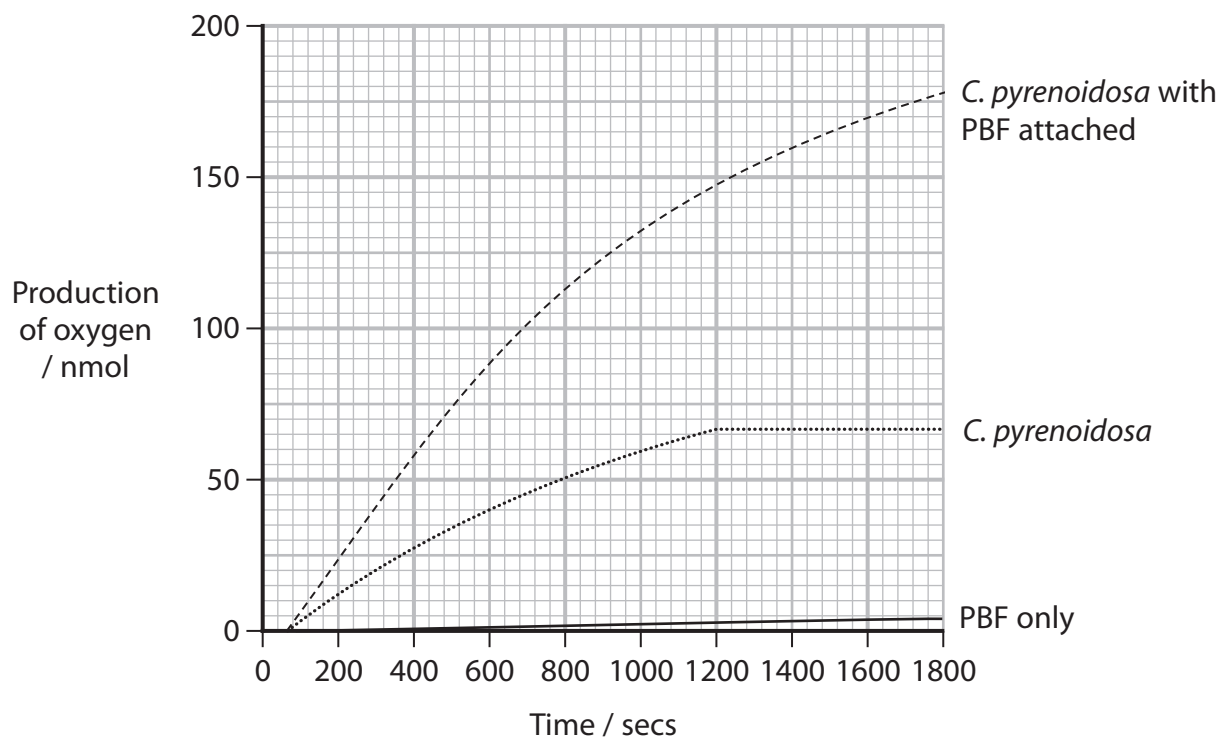


- 5 (a) In an investigation, scientists attached a molecule called PBF to the cell surface of a species of green algae, *C. pyrenoidosa*.

These algae are photosynthetic organisms and PBF absorbs green light.

The scientists measured the production of oxygen by *C. pyrenoidosa*, *C. pyrenoidosa* with PBF attached and PBF only.

The graph shows the results of this investigation.



- (i) Calculate the rate of production of oxygen at 1 200 seconds for *C. pyrenoidosa* with PBF attached.

(3)

Answer nmol secs⁻¹

(ii) Explain the effect PBF has on the production of oxygen.

(3)

(b) These scientists also measured the mass of *C. pyrenoidosa*, with and without PBF attached, after a 20-day period.

They found that the increase in mass of *C. pyrenoidosa* with PBF attached was greater than the increase in mass of *C. pyrenoidosa* without PBF attached.

Explain these findings.

(3)

(Total for Question 5 = 9 marks)